



SAGE UNIVERSITY, BHOPAL, M.P.

SCHOOL OF COMPUTER TECHNOLOGY

A PROJECT REPORT

ON

“AI PLACEMENT PREPARATION AGENT”

B.Tech CSE (Hons.) – Artificial Intelligence

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ABSTRACT

The AI Placement Preparation Agent is an automation-based mini project designed to support students preparing for campus placements. The project uses n8n as the workflow automation platform and integrates Docker, Gemini API, Google Sheets, Google Drive and Gmail.

The system contains two major automation flows. The daily placement workflow retrieves student records from Google Sheets, generates placement-oriented content using Gemini API, stores the generated information and sends the preparation material through Gmail. The generated material includes an aptitude question, a coding question, an interview tip and a motivational quote. A scheduler is configured for automatic execution at 9:00 AM.

The resume feedback workflow is event-driven. When a student uploads a resume PDF to a monitored Google Drive folder, n8n downloads the file, extracts its text and sends the resume to Gemini for analysis. The generated feedback contains an ATS score, strengths, weaknesses, missing skills, suggestions and overall feedback.

The project demonstrates practical use of Generative AI, workflow automation, API integration, cloud-connected productivity services, document processing and automated communication.

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CHAPTER 1 – INTRODUCTION

1.1 Background

Campus placement preparation requires regular practice in aptitude, coding, interviews and resume presentation. Students often need to search for suitable material, organize their preparation and review their progress manually.

The supplied project brief defines an AI-powered Placement Preparation Assistant using n8n, Docker, Gemini API, Google Sheets and Gmail. Its purpose is to automate the generation and delivery of placement material and provide resume feedback when a student uploads a resume.

1.2 Project Motivation

The motivation is to demonstrate how Generative AI and workflow automation can be combined to solve a practical student problem. Rather than building a large standalone application, the project integrates existing services into an automated workflow.

1.3 Project Overview

The completed implementation consists of a daily placement preparation workflow and a resume feedback workflow. The daily workflow is scheduler-driven and communicates with Google Sheets, Gemini API and Gmail. The resume workflow is event-driven through Google Drive and processes uploaded PDF resumes.

1.4 Technologies Used

Technology	Purpose
n8n	Workflow automation and orchestration
Docker	Run n8n locally
Gemini API	AI content generation and resume analysis
Google Sheets	Student data and generated-content storage
Google Drive	Resume upload and trigger source
Gmail	Automated email delivery
GitHub	Project repository and submission

1.5 Learning Outcomes

- Agentic AI concepts
- Workflow automation
- Docker and n8n
- Gemini API integration
- Prompt engineering
- Google Sheets automation
- Gmail automation
- API integration
- Building real-world AI applications

CHAPTER 2 – LITERATURE SURVEY

2.1 Existing Approach

Traditional placement preparation requires students to use separate resources for aptitude, coding, interview preparation and resume review. These resources are useful, but the student remains responsible for deciding what to practice and when to practice.

2.2 Automation and Generative AI Approach

Workflow automation allows multiple services to be connected into triggers and actions. Generative AI can be introduced as an intelligent processing step. In this project, n8n provides orchestration while Gemini API provides generated content and resume analysis.

2.3 Comparative Analysis

Aspect	Manual Approach	AI Placement Agent
Daily content	Manual search	Automatically generated
Scheduling	Student dependent	Automated scheduler
Storage	Scattered	Google Sheets
Email delivery	Manual	Automated Gmail
Resume review	Manual/separate tool	Automated Gemini analysis
Workflow monitoring	Limited	n8n execution history

2.4 Problem Identification

The problem addressed is the repetitive effort needed to organize, generate, store and deliver placement material consistently. The project focuses on automating these repetitive activities while keeping the student responsible for actual learning and preparation.

CHAPTER 3 – OBJECTIVES AND MOTIVATION

3.1 Problem Statement

To design and implement an AI-powered workflow automation system that automatically generates daily placement-preparation content, stores it in Google Sheets, sends it through Gmail, and provides automated resume feedback when a PDF resume is uploaded.

3.2 Objectives

1. Generate a daily aptitude question.
2. Generate a beginner-friendly coding question.
3. Generate an interview tip.
4. Include a motivational quote.
5. Store generated content in Google Sheets.
6. Schedule automatic daily execution at 9:00 AM.
7. Monitor a Google Drive folder for resume uploads.
8. Extract text from uploaded PDF resumes.
9. Analyze resumes using Gemini API.
10. Send placement content and resume feedback by email.
11. Keep API keys and private credentials out of the GitHub repository.

3.3 Scope

The scope includes daily content generation, spreadsheet-based student tracking, Gmail delivery, event-driven resume processing and AI-generated resume feedback. Future versions may include dashboards, authentication, richer analytics and cloud deployment.

CHAPTER 4 – PROPOSED METHODOLOGY

4.1 System Architecture

The architecture connects students and Google services to n8n, with Gemini API acting as the AI processing layer. The daily flow begins with the scheduler and student data, while the resume flow begins with a new PDF in Google Drive.

Layer	Component	Role
Input	Google Sheets / Google Drive	Student data and resume input
Automation	n8n	Triggers, routing and workflow execution
AI	Gemini API	Generation and resume analysis
Storage	Google Sheets	Store student and generated data
Communication	Gmail	Send automated outputs
Runtime	Docker	Run n8n locally

INSERT SCREENSHOT HERE

Figure 1. Overall System Architecture – insert your architecture screenshot here

4.2 Functional Requirements

ID	Requirement
FR1	Read student records from Google Sheets.
FR2	Generate placement content using Gemini API.
FR3	Store generated content in Google Sheets.
FR4	Send daily content through Gmail.

FR5	Execute according to the configured schedule.
FR6	Detect newly uploaded resume PDFs.
FR7	Extract text from resume PDFs.
FR8	Generate structured resume feedback.
FR9	Send resume feedback through email.

4.3 Non-Functional Requirements

- Reliability of workflow execution
- Security of API keys and OAuth credentials
- Maintainability through JSON workflow exports
- Readable email output
- Ability to add multiple students

4.4 Daily Placement Workflow

- Schedule Trigger starts the workflow.
- Student rows are read from Google Sheets.
- Gemini generates placement-preparation content.
- Generated data is stored/updated in Google Sheets.
- Gmail sends the preparation message.

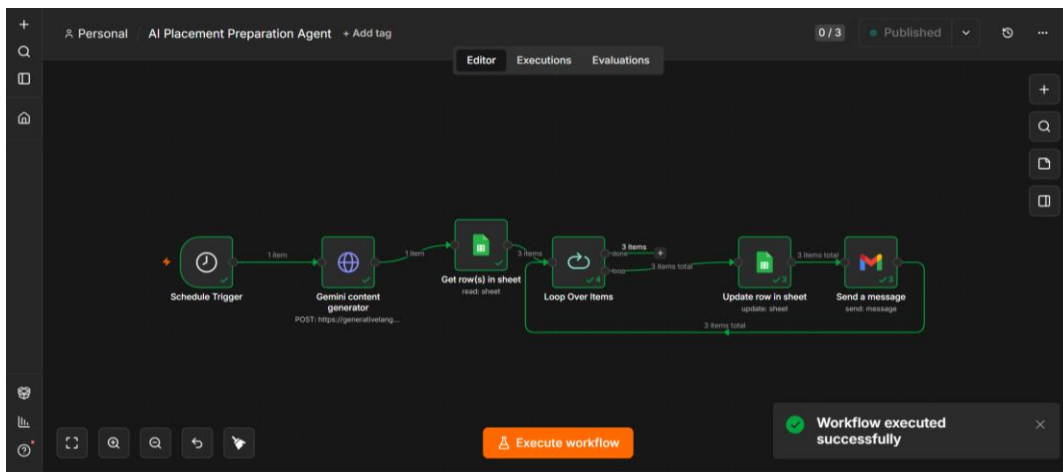


Figure 2. Screenshot of the Daily AI Placement Preparation Agent workflow

4.5 Resume Feedback Workflow

17. Google Drive detects a new resume.
18. n8n downloads the file.
19. PDF text is extracted.
20. Extracted text is sent to Gemini.
21. Gemini returns ATS score and structured feedback.
22. Feedback is sent by Gmail.

INSERT SCREENSHOT HERE

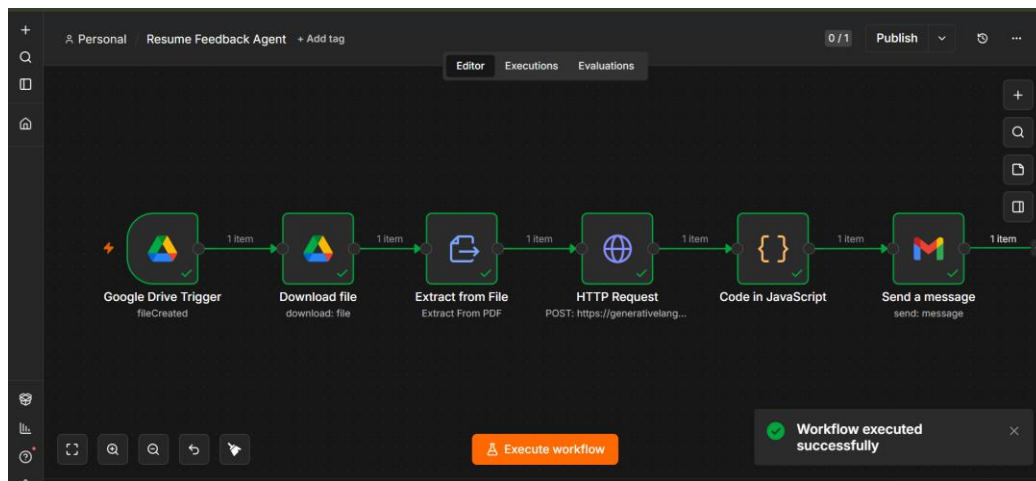


Figure 3. Screenshot of the Resume Feedback Agent workflow

CHAPTER 5 – IMPLEMENTATION AND RESULT ANALYSIS

5.1 Software and Hardware Requirements

Category	Requirement
Operating System	Windows / compatible desktop
Runtime	Docker Desktop
Automation	n8n
AI	Gemini API
Storage	Google Sheets and Google Drive
Email	Gmail
Repository	GitHub
Browser	Modern web browser

5.2 Implementation Details

n8n was run locally using Docker. The workflows were created and tested through the local n8n interface. The daily workflow uses a schedule trigger, Google Sheets, Gemini API and Gmail. The resume workflow uses Google Drive Trigger, Google Drive download, PDF extraction, Gemini API and email delivery.

5.2.1 Google Sheets

Google Sheets is used as the central student-data store. The project brief specifies fields including Name, Email, Date, Aptitude, Coding and Interview Tip. The implemented workflow reads student data and updates the sheet after content generation.

	A	B	C	D
	Name	Email	Date	Aptitude
1				
2	Ishika Sen	ishikasen887@gmail.com	2025-08-10T08:49:59.026-04:00	Question: A sum of money doubles itself in 5 years at simple interest. What is the rate of interest per annum? Answer: Let the Principal be P. Since the money doubles, the Simple Interest (SI) earned is also P. Using the formula $SI = (P * R * T) / 100$: $P = (P * R * 5) / 100$ $1 = R / 20$ $R = 20\%$ Thus, the rate of interest is 20% per annum.
3	manoj sen	senmanoj767@gmail.com	2025-08-10T08:50:02.102-04:00	Question: A sum of money doubles itself in 5 years at simple interest. What is the rate of interest per annum? Answer: Let the Principal be P. Since the money doubles, the Simple Interest (SI) earned is also P. Using the formula $SI = (P * R * T) / 100$: $P = (P * R * 5) / 100$ $1 = R / 20$ $R = 20\%$ Thus, the rate of interest is 20% per annum.
4	ishi sen	ishikasen@gmail.com	2025-08-10T08:50:04.478-04:00	Question: A sum of money doubles itself in 5 years at simple interest. What is the rate of interest per annum? Answer: Let the Principal be P. Since the money doubles, the Simple Interest (SI) earned is also P. Using the formula $SI = (P * R * T) / 100$: $P = (P * R * 5) / 100$ $1 = R / 20$ $R = 20\%$ Thus, the rate of interest is 20% per annum.
5				
6				
7				

Figure 4. Google Sheet showing PlacementAgent student records and generated content

5.2.2 Gemini API

Gemini API is accessed through HTTP Request nodes. Placement prompts generate aptitude, coding and interview content. The resume workflow sends extracted resume text with instructions to return structured feedback.

5.2.3 Gmail Automation

Gmail acts as the communication layer. After processing, the workflow sends the daily preparation material or resume feedback to the student's email address.

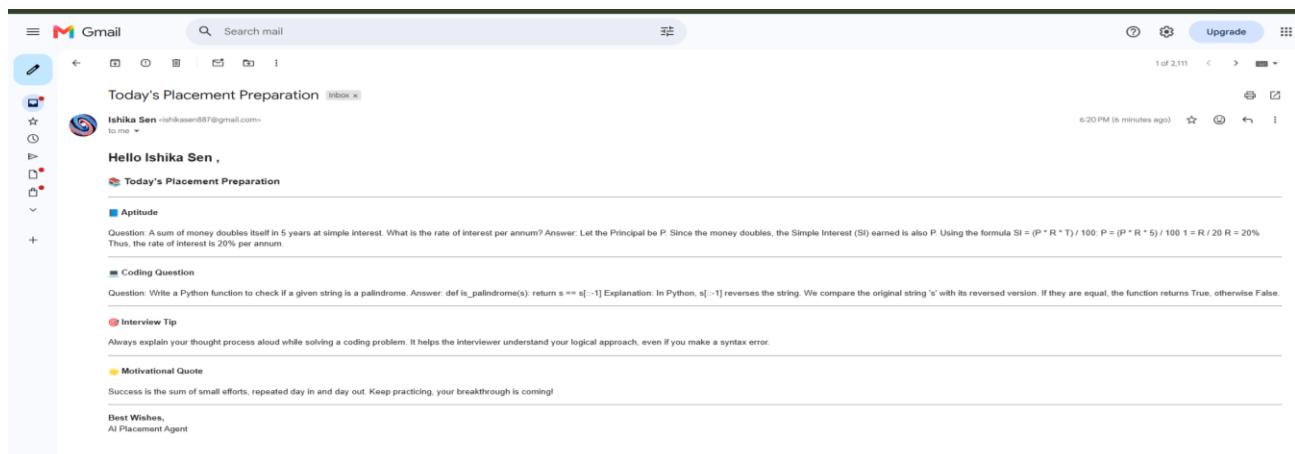


Figure 5. Automated daily placement-preparation email

5.2.4 Resume Feedback Result

During testing, the resume workflow generated an ATS score of 78 together with strengths, weaknesses, missing skills, suggestions and overall feedback. The output identified improvement areas including competitive-coding profile links, live deployment links, cloud/DevOps exposure and system-design basics.

5.3 Result Analysis

Feature	Expected	Observed
Daily generation	Automatic placement material	Working
Google Sheets	Read/update student data	Working
Gmail	Send generated message	Working
Scheduler	Run automatically at 9 AM	Configured
Resume trigger	Detect uploaded PDF	Working
PDF extraction	Extract resume text	Working
AI analysis	Structured resume feedback	Working
GitHub export	Workflow JSON without secrets	Prepared

5.4 Project Scheduling

Phase	Activity	Status
1	Docker and n8n setup	Completed
2	Daily placement workflow	Completed
3	Gemini integration	Completed
4	Sheets and Gmail integration	Completed
5	Resume feedback workflow	Completed
6	Testing and credential reconnection	Completed
7	Secret removal for GitHub	Completed
8	Documentation and submission	Completed / Finalizing

5.5 Software Testing

Test Case	Action	Expected	Result
TC01	Execute daily workflow	Content generated	Pass
TC02	Read Google Sheet row	Student record returned	Pass
TC03	Send Gmail	Email delivered	Pass
TC04	Upload resume PDF	Workflow triggered	Pass
TC05	Extract PDF	Text available	Pass
TC06	Gemini resume analysis	Structured result	Pass
TC07	Update Google Sheet	Data stored	Pass
TC08	Publish schedule	Workflow active	Pass

CHAPTER 6 – CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The AI Placement Preparation Agent demonstrates how workflow automation and Generative AI can be combined to create a practical placement-support system. The project automates repetitive activities such as generating preparation content, storing it, sending it by email and analyzing uploaded resumes.

n8n provides a visual automation layer, Docker provides the local runtime, Gemini API provides AI capabilities, Google Sheets provides structured storage, Google Drive provides the resume event source and Gmail provides automated communication.

The workflows were tested with the configured integrations. The daily workflow was configured for scheduled execution and the resume workflow was tested with PDF extraction and AI-generated feedback. API secrets were removed from the files intended for GitHub submission.

6.2 Future Scope

- Student dashboard for preparation history.
- Authentication and role-based access.
- Difficulty-based aptitude and coding generation.
- LeetCode/HackerRank profile integration.
- Resume improvement tracking.
- Mock interview automation.
- Secure cloud deployment.
- Database-backed analytics.
- Additional notification channels.
- AI output quality monitoring and evaluation.

6.3 Limitations

- The project depends on third-party APIs and Google services.
- AI resume feedback is guidance and not a guaranteed ATS decision.
- Google Sheets is suitable for this mini project but not intended as a high-scale database.
- Imported workflows require credentials to be configured again in another n8n instance.

/ BIBLIOGRAPHY

- [1] AI Placement Preparation Agent project brief supplied for the mini project.
- [2] n8n workflow automation and node documentation.
- [3] Google Gemini API / Google AI Studio developer documentation.
- [4] Google Sheets and Google Drive integration documentation.
- [5] Gmail API and n8n Gmail node documentation.
- [6] Docker documentation for containerized application execution.
- [7] GitHub documentation for repository management and secret scanning.

GITHUB SUBMISSION CHECKLIST

- README.md – project overview, setup and workflow explanation.
- workflow.json – exported workflow without API keys or private credentials.
- workflows/ – Daily Placement Agent and Resume Feedback Agent JSON exports.
- docker-compose.yml – Docker configuration for local n8n setup.
- prompt.txt – non-secret prompts used by the workflows.
- screenshots/ – screenshots of both workflows and important outputs.
- report.pdf – final exported report.

The project brief recommends a Placement-Agent folder containing docker-compose.yml, README.md, workflow.json, prompt.txt, GoogleSheet/, screenshots/ and report.pdf. Before GitHub upload, API keys, OAuth tokens, service-account secrets and private credentials must be removed.